

UTCr README
(12 February 2014)

THIS DIRECTORY CONTAINS:

- Results of the computation of UTCr.
- Data from clocks used in UTCr calculation
- Documents with information related to UTCr, including guidelines for participation.

1. RESULTS

- Official results published from 1 July 2013 are in:
 - files UTCr_yyww with daily values of [UTCr-UTC(k)] at 0 h UTC for year yy and week ww starting from UTCr_1326. They are also available at [http://www.bipm.org/jsp/en/TimeFtp.jsp?TypePub=publication](http://www.bipm.org/jsp/en/TimeFtp.jsp?TypePub=publication;);
 - files UTCr-lab contain daily values of the differences between UTCr and UTC(lab) at 0 h UTC starting from MJD 56467. They are also available at <http://www.bipm.org/jsp/en/TimeFtp.jsp?TypePub=publication>.

The results are also available at <ftp://tai.bipm.org/UTCr/Results/>.

- Sub-directory "pilot_experiment" contains the results published during the pilot experiment phase:
 - files UTCr_yyww with daily values of [UTCr-UTC(k)] at 0 h UTC for year yy and week ww from UTCr_1205 to UTCr_1325;
 - files UTCr_p-lab with daily values of the differences between UTCr and UTC(lab) at 0 h UTC from MJD 55956 to MJD 56466.

2. DATA

Clock data used for UTCr_yyww calculation can be found in **/Data/YYYY/clocks/** where YYYY stands for the year corresponding to 'yyww'. This directory contains daily clock data formatted for UTCr computation:

- Files 'uh____dd.ddd' are daily values UTC(k) - clock of all clocks for MJD dddd.

The file format is:

```
0000000001111111111222222222233333333333444444444455555555556
123456789012345678901234567890123456789012345678901234567890
XXXXX.XXX  100YY CCCCCC aaaaaaa.aaa          AAAA
```

Description of a line containing clock data:

field 01 to 09 : XXXXX.XXX is MJD in format f9.3.
field 10 to 12 : blank characters
field 13 to 17 : 100YY is the laboratory code
field 18 : blank character
field 19 to 25 : CCCCCC is the code of one clock on 7 digits
field 26 : blank character
field 27 to 37 : aaaaaaa.a is [UTC(lab)-clock] in format F11.3, in ns
field 38 : blank character
field 39 to 48 : 10 characters field reserved for future use
field 49 : blank character
field 50 to 53 : AAAA is the laboratory acronym, up to 4 characters

- Files '**shyyw**____' are declared clock steps used for UTCr_yyww calculation.

The file format is:

0000000001111111111222222222233333333334444444444555
1234567890123456789012345678901234567890123456789012
XXXXX.XXX CCCCCC cccccc.ccc bbbbbb.bbb AAAA 100YY

Description of a line containing clock step data:

field 01 to 09 : XXXXX.XXX is MJD on 5 digits and the fraction of day of the step
field 10 : blank character
field 11 to 17 : CCCCCC is the code of the clock on 7 digits
field 18 : blank character
field 19 to 29 : cccccc.ccc is the amount of the time step in format F11.3, in ns
field 30 : blank character
field 31 to 41 : bbbbbb.bbb is the amount of the frequency step in format F11.3, in ns/d
field 42 : blank character
field 43 to 46 : AAAA is the laboratory acronym, up to 4 characters
field 47 : blank character
field 48 to 52 : 100YY is the laboratory code on 5 digits

2. DOCUMENTS

- Files FA-TA.452/453/454 are the messages circulated among the laboratories informing on the pilot experiment on UTCr.
- Files FA-TA.xxx dated from 1 July 2013 concern the regular calculation of UTCr.
- Files UTCr-Bull_xx are the information bulletins on UTCr issued by the Time Department.
- File ReadMe_guidelines.pdf gives the information on data submission for UTCr via FTP server.